

Rise of the Autonomous Machines

Shaoshan Liu¹ *Senior Member IEEE*

Jean-Luc Gaudiot² *Life Fellow IEEE*

¹PerceptIn Inc

²University of California, Irvine, U.S.A.

Abstract:

After decades of uninterrupted progress and growth, information technology has so evolved that it can be said we are entering the age of autonomous machines, but there exist many roadblocks in the way of making this a reality. In this article, we make a preliminary attempt at recognizing and categorizing the technical and non-technical challenges of autonomous machines; for each of the ten areas we have identified, we review current status, roadblocks, and potential research directions. It is hoped that this will help the community define clear, effective, and more formal development goalposts for the future.

The Age of Autonomous Machines: The Sixth Layer of Information Technology

As an increasing number and kinds of autonomous machines enter our daily life, the age of autonomous machines is upon us and a whole new era of information technology begins. How did we get here? Before delving into autonomous machines, let us first review the evolution of computing machinery. Information technology took off in the 1960s when Fairchild Semiconductors and Intel established its foundation by producing the first silicon microprocessors, with the attendant explosive growth of the Silicon Valley (See Figure 1). At first, although microprocessor technologies greatly improved industrial productivity, the general public had limited access to it. This changed in the 1980s, with the advent of personal computers, and later Apple Macintosh and Microsoft Windows, using a Graphical User Interface (GUI). The second layer had been laid, and the vision of ubiquitous computers, targeted for use by untrained personnel at home, had become a possibility.

With virtually everyone having access to computing power by the early 2000s, Yahoo and Google laid the third layer, connecting people—indirectly, with information available through search engines. With the third layer, the internet, a core infrastructure to enable later layers, could become ubiquitous.

Beginning with Facebook in 2004, social networking sites created the fourth layer of information technology by allowing people to directly connect with each other, effectively moving the whole of human society to the World Wide Web.